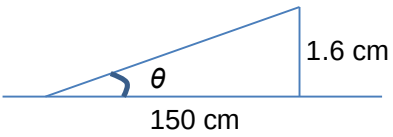


	Some students thought they could just sum up the individual intensities. Some thought that O is a point of destructive interference.	
4(a)(i)	Path difference = 2×0.020 $= 0.040 \text{ m}$ Path difference = $\frac{\lambda}{2}$ $\lambda = 2 \times 0.040$ $= 0.080 \text{ m}$	[1] for correct path difference [1] for correct wavelength
4(a)(ii)	The sound waves from path LXM and LYM travel in <u>opposite directions</u> and <u>meet</u> . Since both waves are of <u>equal amplitude, frequency and speed</u> , they <u>superpose</u> to form a stationary wave.	[1] [1]
4(b)(i)	 $\theta = \tan^{-1}\left(\frac{1.6}{150}\right) \quad \theta = 0.61^\circ$ angle between the two first orders = 2θ $= 2 \tan^{-1}\left(\frac{1.6}{150}\right)$ $= 1.2^\circ$	[1] for correct θ [1] for correct answer
4(b)(ii)	$d \sin \theta = n\lambda$ $d \sin 0.61^\circ = 1(600 \times 10^{-9})$ $d = 5.6253 \times 10^{-5} \text{ m}$	[1] for correct d

	$\begin{aligned} \text{number of nylon threads per mm} &= \frac{1 \times 10^{-3}}{d} \\ &= \frac{1 \times 10^{-3}}{5.6253 \times 10^{-5}} \\ &= 17.8 \\ &\approx 18 \end{aligned}$	[1] for correct answer
5(a)	Electric field strength at a point is the electric force per unit positive	[1]